

	Course Work	DOC	BAC.ACAD.PM.03.F02.V3.0
		EFF	01 MAR 2023
		V	3.0

School of Computing and Information Systems
PROGRAMME : BSc Computer Systems Engineering

Academic Year: Jan – Jun 2026

Level: Year 1

Module Name: MT103 – Computer programming with JAVA

Module Code: MT103

Coursework No: 1

Coursework type: Assignment/Report/Presentation

Module Leader: *MP, PS, SM, TNA, BL*

Date: 03 March 2026

Submission Date: 18-May 2026

BAC.ACAD.PM.03. F02.V3.0

Instructions to candidates

1. Candidates must attempt **ALL** questions.
2. You are to make your submission on turn-it-in. You may consult with your tutor/lecturer on how this will be done.
3. Ensure that you have an account on turn-it-in by going to **www.turnitin.com**. Use the credentials provided for accessing this system. If you do not have them, get hold of the tutor/lecturer as soon as possible.
4. Any work with plagiarism level above **30 % will not be marked**. It your responsibility to make sure that your plagiarism level is within this level. Monitor it on regular bases. If your share your solution with others, chances of the plagiarism rising above this level are high.
5. It is your responsibility to ensure that you have **Computer Programming with JAVA** module in turn-it-in before submission date and you do not drop the module. Consult with your tutor/lecturer if this is not the case.

LEARNING OUTCOME (S) TO BE ASSESSED

By completing this assignment students should be able to:

- LO1- Develop a basic GUI application in Java
- LO2 - Use event-driven programming
- LO3 - Apply conditional statements
- LO4 - Perform arithmetic calculations
- LO5 - Structure programs using methods
- LO6 -The use of I/O file streams.

ASSIGNMENT BRIEF

Assignment Objective

The purpose of this assignment is to assess your understanding of:

- Writing a complete Java program
- Using variables and data types
- Accepting user input
- Using decision-making statements
- Using loops
- Creating and calling methods
- Reading and writing to text files

The assignment scenario

SmartCare Private Clinic is a growing healthcare facility located in Francistown. The clinic provides:

- General medical consultations
- Specialist consultations
- Emergency medical services
- Basic laboratory testing

Currently, the clinic uses a **manual paper-based billing system**. After each consultation, the receptionist calculates the patient's bill using a calculator and records the details in a physical ledger book.

Problems With the Current System

The clinic management has identified several issues:

- 1) Manual calculations sometimes result in errors.
- 2) Patient records are difficult to retrieve when needed.
- 3) Searching for old records takes too much time.
- 4) Paper files are easily damaged or misplaced.
- 5) There is no digital backup of billing information.

Because of these challenges, the clinic has decided to computerize its billing system.

Your Role

You have recently been employed as a Junior Software Developer at a small IT company contracted to develop this system.

Your responsibility is to:

Plan, design, and create a GUI-based Patient Billing System using Java that will improve efficiency and reduce errors at the clinic.

System requirements

The clinic manager has specified that the system must:

1) Capture Patient Information

- Patient Name
- Patient ID
- Age
- Gender

2) Allow Selection of Consultation Type

Only one consultation type can be selected per visit:

- General Checkup (BWP 300)
- Specialist Visit (BWP 500)
- Emergency Visit (BWP 1000)

3) Allow Selection of Laboratory Tests

During a visit, a patient may require additional laboratory tests such as:

- Blood Test
- X-Ray
- COVID Test
- Each test costs BWP 150.

4) Automatically Calculate the Bill

The system must:

- Calculate Subtotal
- Add VAT (14%)
- Display the Final Total

This ensures accuracy and consistency.

5) Store Billing Records Digitally

The clinic requires that:

- Every transaction must be saved in a file called **clinic_records.txt**.
- Records must not overwrite previous data.
- The system must maintain historical billing records for auditing purposes

6) Provide Search Functionality

The clinic receptionist must be able to:

- Enter a Patient ID
- Retrieve that patient's billing record from the file

This feature will significantly reduce time spent searching through physical file

Design Requirements.

OOP Practices – modularize the program through separation of classes, where possible application of inheritance, polymorphism, interfaces.

Write a brief report about your system including introduction, classes used and methods, testing, challenges and future improvements of the system.

Guidelines

- 1) Implement your program with a GUI
- 2) Your programming method should be Object Oriented i.e. good programming practices such as separation of classes.
- 3) Apply OO principles such as abstraction, inheritance, polymorphism and encapsulation.
- 4) Provide test cases for your program and show sample input/output outcome.
- 5) Students must use their own datasets, **sharing data files will attract serious penalties including marking the entire system to 0 marks.**
- 6) A drag-and-drop editor may be used when designing the GUI.
NOTE: The student owns the assignment 100%, including any auto-generated code produced by the editor. Therefore, the student must be able to present, explain, and defend the entire code during presentation.

Marking Template

Marks (100%)



Student Name: _____

Student ID : _____

A)	OOP Practices – modularization of program through separation of classes, where possible application of inheritance, polymorphism, interfaces. Proper implementation of Encapsulation and abstraction concepts will attract marks. Use of different classes to tackle different tasks Superclass/subclass relationships and or interfaces Use of access modifiers 4 marks 3 marks 3 marks	Allocated marks	Awarded marks
		10 Marks	
B)	A menu to select an operation 2 marks for displays of menu, 2 marks keyboard input, 2 marks for determining option chosen, 2 marks for loop , 2 marks for error message if wrong option input	Allocated marks	Awarded marks
		10 Marks	
C)	Ability to add a Patients to an array with a maximum of 20 records. 2 marks for array declaration of array 2 marks for keyboard entry of employee details 2 marks Patient object creation 2 marks object storage in array 2 marks array index increment	Allocated marks	Awarded marks
		10 Marks	
D)	Ability to show all Patient details added to the array. 1 mark for loop start 1 mark loop end 1 loop increment 2 marks employee detail display	Allocated marks	Awarded marks
		5 Marks	

E)	Ability to search for Patient of a chosen title. 2 marks for input of Patient id or names 2 marks loop start 2 marks loop end 2 marks loop increment 2 marks comparison of name or id and display of Patient details	Allocated marks	Awarded marks
		10 Marks	
F)	Ability to display only male or female Patient at a particular time. 2 marks input of male or female 2 marks loop start. 2 marks loop end 2 marks loop increment 2 marks check for gender and displays	Allocated marks	Awarded marks
		10 Marks	
G)	Ability to display all Patient and amounts they owe the clinic 2 marks loop start 2 marks loop end 2 marks index increment 2 marks gross salaries calculation 2 marks display of employee details and gross salary.	Allocated marks	Awarded marks
		10 Marks	
H)	the system should allow a search for an Patient and display all laborotary the performed. 1 mark input of employee id or names 1 for loop control 2 mark for comparison of input details and array data 1 mark for display of employee details and calculated salary details	Allocated marks	Awarded marks
		5 Marks	

I)	The system should have the ability to display the total number of patients who performed a particular consultation type.	Allocated marks	Awarded marks
	3 mark for obtaining a list of patients per consultation type 1 loop control 1 count per type	5 Marks	
J)	A brief report on your system including introduction, purpose, status report, 10 future development, test cases and conclusion	Allocated marks	Awarded marks
	2 marks introduction 2 marks purpose of the application 2 marks per statement max 3 identification what works (6 Marks max) 2 marks per statement max 3 identification what doesn't work(6 Marks max) 2 marks future developments 1 mark per item 6 marks, 1 mark per test case 1 mark conclusion	25 Marks	
Total		100 mark	

Appendix 1: Assignment submission cover sheet

ASSIGNMENT SUBMISSION COVER SHEET

Student Id:

Student names:

Student email:

Cohort:

Assignment title:

Date of submission:

Programme of Study:

Year of Study:

Intellectual property statement

By checking the box below, I certify that this assignment is my own work and is free from plagiarism. I understand that the assignment may be checked for plagiarism by electronic or other means and may be transferred and stored in a database for the purposes of data-matching to help detect plagiarism. The assignment has not previously been submitted for assessment in any other unit or to any other institution. I

have read and understood the Botswana Accountancy College plagiarism guidelines policy.

☒ Agree

Signature.....

Date.....19 MAY 2026